

SSGMCE	SHRI SANT GAJANAN MAHARAJ COLLEGE OF ENGG., SHEGAON - 444 203.		LABORATORY MANUAL	
	PRACTICAL EXPERIMENT INSTRUCTION SHEET			
	EXPERIMENT TITLE:	TO DECLARE VARIABLE, EXPRESSION, FUNCTION AND EXECUTE R SCRIPT.		
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SUBJECT: BIG DATA ANALYTICS			SEMESTER: VI	PAGE: 1 OF 3

Aim: To declare variable, expression, function and execute R script.

Facilities: Computer with RStudio installed.

Theory: R is a programming language mostly used for statistical computing and graphical representation to analyze and visualize data.

Basic syntax:

To output *text*, use single or double quotes, while for *numbers*, type the number without quotes.

Ex: "Hello World!", 5

Basic mathematical *operations* can be done directly.

Ex: 2 + 3

Printing can be done directly like mentioned above, still *print()* function is available for use. In some cases *print()* function is necessary, like in *loops*.

Ex: `print("Hello World!")`

To add *comment*, use # followed by any text.

Ex: # This is a comment

Operators:

In R, operators are divided into five major categories:

1. Arithmetic operators (+, -, *, /, %%, %/%, ^)
2. Relational/Comparison operators (>, <, ==, >=, <=, !=)
3. Logical operators (&, |, !, &&, ||)

4. Assignment operators (<-, =, <<-, >-, >>=)

5. Miscellaneous operators (:, %in%, %*%)

Variables:

A variable provides us with named storage that our programs can manipulate. A valid variable name consists of letters, numbers and the dot or underline characters. The variable name starts with a letter or the dot not followed by a number. Below table gives clear idea about legal and illegal variable names, variable names are case sensitive.

Variable Name	Validity	Reason
var_name2.	Valid	Has letters, numbers, dot and underscore
var_name2%	Invalid	Has the character '%'. Only dot(.) and underscore allowed.
2var_name	Invalid	Starts with a number
.var_name, var.name	Valid	Can start with a dot(.) but the dot(.) should not be followed by a number.
.2var_name	Invalid	The starting dot is followed by a number making it invalid
_var_name	Invalid	Starts with _ which is not valid

R does not have a command for declaring a variable. A variable is created the moment we first assign a value to it. To assign a value to a variable, use the <- or = sign (However, <- is preferred in most cases because the = operator can be forbidden in some context in R.). To output (or print) the variable value, just type the variable name:

Ex: a<-5

Data Types:

Variables can store data of different types, R has a variety of data types and object classes. Here variables do not need to be declared with any particular type. Some of the basic data types are *numeric*, *integer*, *complex*, *character*, *logical*, etc. With *class()* function we can identify data type of any variable.

If:

Else If:

If Else:

Nested If:

Switch Statement:

While Loop:

For Loop:

Loop Control Statement: Break, Next Statements

Function Creation:

Function Call:

Return Function:

Built-in math functions:

With these functions, we can perform mathematical operations on numbers. Some of these functions are min(), max(), sqrt(), abs(), ceiling(), floor(), etc.

String functions:

nchar(): gives number of characters in a string.

grep(): checks whether a character or a sequence of characters are present in a string.

paste(): concatenate two strings.

Creation and execution of R-script:

Conclusion: We have successfully performed various basic operations in R.

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